

Faculty of Engineering and Design Te Herenga Auaha



Department of Engineering Science and Biomedical Engineering Te Kura Pūtaiao Pūhanga

Part 4 Projects Handbook

Latest Revision: 02 March 2026

1. Introduction

The Part 4 Projects (P4P) provide an opportunity for students to work largely on their own initiative, under supervision, on a topic of interest to Engineering Science and Biomedical Engineering students. Work takes place over a complete academic year, requiring students to enrol in both ENGSCI 700A/B. Students will work in pairs under the direction and guidance of a project supervisor (and/or a co-supervisor). The research project requires independent thought and action. We hope that students will feel a personal sense of achievement.

The focus of projects will vary by department and specialisation, ranging from experimental, lab, field, computational, and analytical work. **Individual grades are awarded for this research project**, which is assessed solely on the **Final Research Report**. ENGSCI 700 A/B “Research Project” meets all the requirements of an NQF Level 9 course. The 2x15-point course (total of 30 points) spanning Semesters 1 and 2 is a required course for the BE(Hons) programmes.

Work on the project commences in March and continues through October, with the expectation that students will spend, on average, about 10 hours per week on their project throughout this period. It is important that all projects start at the same time so that no student group gains an advantage.

Students will need to draw upon a good level of theoretical knowledge and the skills they have acquired so far in the degree. In some areas, students will need to extend their knowledge and expertise.

The research will start with the preparation of the research objectives and project scope, and a literature review to identify prior work and explore alternative approaches to solving the problem. This will lead to the detailed work on a solution. The project should include a research component to help students develop their research skills.

2. Learning Outcomes

At the end of this course, our students should be able to:

- i. Retrieve, assess, and evaluate existing research outcomes and technologies relevant to the field of research. Demonstrate an in-depth understanding of the related work, including an in-depth knowledge of the literature, the well-known authors in the research field, the related terminology, and the research findings (theories, models, structures, designs, principles etc.).
- ii. Classify, summarise, explain, and critique the basic findings of the literature review and identify gaps in the current knowledge. Demonstrate an ability to extract and list the key issues from the literature review in order to conclude to the required problem statements and research questions. Demonstrate the ability to synthesise an exemplary problem statement, propose research questions, and formulate research hypotheses.
- iii. Demonstrate clearly their knowledge of the underlying theory, methods, and procedures. Evaluate and compare techniques quantitatively. Demonstrate an in-depth understanding of the required experiments and/or simulations and how they actually relate to the research questions or hypotheses. Demonstrate an understanding of the experimental design or numerical study, giving consideration to statistical significance and quality of collected data.
- iv. Correctly use the theory and research methods and efficiently conduct the required experiments and/or simulations. Demonstrate the ability to use data to extract appropriate models, cluster data into relevant groups, extract major and minor components, perform statistical analysis, organise data into appropriate tables, graphs, and diagrams, interpret data and compare them with previously published data, address the validity or inappropriateness of the data (discussing possible limitations), critique and recommend future improvements for the experimental design and the research direction.
- v. Design, develop, evaluate, and experimentally validate appropriate solutions for the selected research questions and problems. Demonstrate proficiency related to hardware and software co-design (where applicable).
- vi. Demonstrate their proficiency in engineering/technical writing by being able to generate a technical research report summarising the research findings and to create a research compendium containing all the required supplementary information.
- vii. Deliver a public presentation and an exhibit (project display) to an audience with broad background knowledge about the findings of the research project.
- viii. Provide knowledgeable and logical explanations and expert opinions in relation to the research findings. Demonstrate the ability to add, at a satisfactory level, to the existing knowledge, articulating the importance of the work, in relation to solving a larger, 'real-world' problem.

3. Key Components and Dates

The various research components are shown in Table 1.

Table 1. P4P Components and Important Dates.

P4P Component	Type	Date
Project Scope, Research Objectives, & Literature Review	Compulsory (<u>individual submission</u>)	Thursday, 2 April 2026 11:59 pm
Mid-Year Technical Report	Compulsory (<u>individual submission</u>)	Friday, 24 July 2026 11:59 pm
Conference Presentation	Compulsory (<u>participation in event</u>)	Friday 31 July 2026 (Whole day)
P4P Display Day Poster	Compulsory Presentation Deliverable of P4P Display Day (<u>joint submission</u>)	Monday, 12 Oct 2026 11:59 pm
Final Research Report	Assessed / Compulsory (100% of grade) (<u>individual submission</u>)	Sunday, 18 Oct 2026 11:59 pm
Research Compendium	Compulsory (<u>joint submission</u>)	Tuesday, 20 Oct 2026 11:59 pm
P4P Display Day	Compulsory (<u>participation in event</u>)	Thursday, 22 Oct 2026

The passing grade for this course is solely determined by a written research report prepared independently by each student, supported by an online technical appendix (termed a research compendium). This course also requires students to complete a number of compulsories, non-graded components, as outlined above. **Failure to deliver the compulsory components may result in a DNC ("Did Not Complete") grade for this course.**

***Unless special prior approval has been provided by the course coordinator.**

4. Staff and Student Involvement

Staff-Student Interaction

Meetings between staff and student pairs are an essential part of the learning and research training process. Students are expected to lead the meetings, take notes, record the issues discussed and the actions decided, propose targets, and report on their progress. The expectation is for a weekly meeting with each project pair. However, different meeting cadences can be agreed upon.

Professional conduct is expected of all parties. All emails about the project progress should include the project partner. Failure to attend meetings should be recorded; if it occurs more than once without an excuse, the supervisor should inform the student by email (cc-ing the P4P coordinator) that this is unacceptable.

Student's Role

Although supervisors propose the research project, the project's "ownership" transfers to the student once it begins in semester 1. Students must not rely on supervisor, technicians or other people to perform experiments or carry out project work on their behalf. The responsibility for conducting experiments, collecting data, completing results and analysis, and undertaking all project task rests with the student.

Students are expected to spend **at least 10 hours a week on their project** throughout the academic year. The academics should ensure the project's workload fits within this time frame. The work within the project scope should be appropriately divided. When it is not possible to assign separate roles, students will work jointly, preparing individual reports.

Supervisor's Role

Supervisors should ensure that projects do not start early, as an early start for one project is not equitable to other students.

Supervisors are expected to provide specific and constructive feedback on the student's research progress. They should also monitor their students' progress. If students are falling behind, missing meetings, or are not adequately contactable, the supervisor should notify them by email, also copying the project coordinator. Supervisors should always remember that the project "ownership" belongs to their students. They should not take it personally if their students don't do a good job.

Although **students should consult their supervisors** regarding the organisation and content of reports, **supervisors are not permitted to make substantial editorial changes** and they are expected to provide limited feedback on the report writing (e.g., feedback on abstract), as well as high-level feedback on the report structure and the content. It is permissible for the supervisors to comment on a small section (e.g. a single page or the abstract) so as to provide an example of how the writing style of the students can be improved.

Technicians' Role

The technicians in each lab will assist students in achieving their research goals by offering expert guidance on prototype design, experimental setup preparation, experiment execution, equipment use, etc. The technicians can also assist students with ordering the materials and goods required for the successful completion of the project.

Purchasing

Students need to coordinate with the lab technician for purchasing the required consumables or desired equipment. Purchase should be approved (via email) by the supervisor.

Students are strongly discouraged from purchasing with their own funds. It is against procurement procedures and is difficult and time-consuming.

5. Project Components: Instructions

Project Scope, Research Objectives, and Literature Review Report

The first submission (report) should focus on the project scope, objectives, and the literature review. **The project scope and objectives part should be 1-2 pages long**, while the **literature review part should be 5-6 pages long**. It should include **a project schedule / plan in the form of a Gantt chart**.

The goal of this report is to describe the research that the students will be doing for their project. The students should present the problem that they are trying to solve, the important research question(s), the objectives, and explain why this research is important. The purpose of the literature review is twofold:

- to demonstrate that students are aware of the research relevant to their project
- to summarise the important information or findings that will form the foundations of the work they are intending to carry out.

The review should also show the breadth of the research area and some depth in particular areas. Students can use the final research report template to prepare this deliverable.

Each report (digital file) will be submitted to TurnItIn for plagiarism check (see section 7 for more details). Students should make sure that the .pdf file submitted can be read by TurnItIn and that it is editable.

ACADEMIC DISHONESTY WILL NOT BE TOLERATED.

Mid-Year Technical Report

The **mid-year technical report should be 4-6 pages long** (excluding title page, Table of Contents/List of Figures/List of Tables, Abstract, References, and Appendices). The content of the mid-year technical report depends on the project nature and scope. The students are expected to consult their supervisors in order to guarantee that they will submit an appropriate report with appropriate content. The project scope, objectives, and literature review should **NOT** be included in the mid-year technical report. Students can use the final research report template for preparing this deliverable (available on Canvas).

Final Research Report

The final research report will typically be **12,000 words in length, and must be no shorter than 8,000 words, and no longer than 13,000 words**. **The report must also not exceed a total of 30 pages in length**, where both the number of words and pages should be counted from the *Introduction* to the *Future Work* section inclusive (see below). **All reports must include at least 8,000 words**. The font should be Times New Roman, and the **font-size 12 pt**.

A final research report template will be provided. The report must be made up of the following sections: Title Page, Abstract, Signed Statement of Contribution,

Acknowledgements, Table of Contents, Glossary of Terms, Introduction*, Literature Review*, Middle sections* appropriate to your project work, Discussion*, Conclusions*, Future Work*, References, and Appendices. All sections marked * are included in both the word count and page count.

Further technical material that supports the research findings will be provided as a research compendium that forms an online appendix to the report. The same research compendium can be submitted by both students.

No acknowledgement of AI use is required in the research report.

Conference Slides and Presentation

Appropriate templates will be made available through the Canvas page of ENGSCI700A/B. The conference sessions will start at 8:00 am and they will be 120 minutes long. Each group will have 7-8 minutes (+ 2 min for QA). A detailed conference programme will be available later during the year. The presentation slides should be prepared in 4:3 aspect ratio. **Please note that this is considered a public event and no confidential information should be included in the slides.** More instructions about how to prepare a proper presentation can be found at:

<https://learningessentials.auckland.ac.nz/presenting-your-work/>

Display Day Poster

The display day poster should be prepared in landscape orientation and will be printed on A1-sized paper. The display day presentation will take place in building 405. A detailed programme for the Display Day and an annotated floor map showing the exact location of each project will be announced later this year. **Please note that the P4P Display Day is considered a public event, and no confidential information should be included in the poster.**

Research Compendium

The research compendium should provide all the supporting materials for the project report. It should provide information about all undertaken research (e.g., for aspects of research that are not included in the report), details on the experiments, tests, and procedures (including experimental setups, equipment used, serial numbers, calibration, test conditions, special components, etc.), data that was used for analysis or to generate plots for the report, CAD files, images, video and/or audio files. The research compendium should also contain a ReadMe file that will contain a detailed description of the structure, organisation, and contents.

6. Component Quality Expectations

Final Report (Compulsory)

Expectations: Clarity, professional structure, technical descriptions that convey understanding of the field, analysis of results, excellent English, technical accuracy, sufficient technical depth, conciseness and completeness, use of sufficient and relevant references.

Conference Presentation (Compulsory)

Expectations: Clarity of presentation, appropriate structure, accurate technical content, sufficient technical depth, appropriate question handling, professional presentation style and skill, descriptions that convey understanding of the field of research, focus on concepts over unnecessary details.

Display Day Poster Presentation (Compulsory)

Expectations: Clarity of presentation, good flow and good visual linking, good readability of poster content, use of appropriate diagrams, graphs and/or pictures, use of appropriate technical content that conveys essential components and outcomes of the research, sufficient technical depth, visual appeal: use of aesthetically pleasing colours, pictures, background etc.

Research Compendium (Compulsory)

Expectations: The research compendium forms an online technical appendix to the report. A well-presented compendium should contain sound information and materials that will allow replication of the research project work and outcomes by future students and/or researchers. A good research compendium can be easily read, followed, understood, and further developed.

Assessment and Marking

The research report will be assessed in accordance with the University of Auckland procedures as below link:

<https://www.auckland.ac.nz/en/about-us/about-the-university/policy-hub/education-student-experience/sub-doctoral-research/examination-sub-doctoral-postgraduate-research-components.html>

A rubric will be provided to the academic staff members (examiners and assessors) and students to assist them with their assessment and marking.

7. Notes on Plagiarism

With regard to the individual nature of the project report and plagiarism issues that could arise, the following should be noted:

- Each individual student's report must show that it has been written independently of their partner's report.
- A declaration (available in the final report template) has to be completed by all students, signed off by the student, and appear in the report, stating clearly what has been developed during the project and what was available from previous years (e.g. previous P4P).
- TurnItIn is used to compare reports and derive a similarity score for each report.
- If the examiner and assessor judge that the text and/or structure of project reports show collaboration in terms of a lack of independent writing, analysis, or significant overlap in written material, the student will be severely penalised for plagiarism.
- Students are advised/encouraged to use TurnItIn to check that their reports have low similarity scores.

The university regulations with respect to plagiarism are very strict and the penalties can be very severe (coursework penalties and severe fines). Students are very strongly advised not to share digital files or hard copies of written text.

Structure of the report: It is permissible that the two reports of a project have the same overall structure as dictated by the project itself. Beyond this, the reports must show no evidence of collusion.

Illustrations: It may be inevitable that the illustrations in reports of project partners are very similar and this is allowed. If a figure that is used has been solely produced by one of the partners, the figure's caption should include a formal reference to the partner's project report.

8. Notes on using AI

The Part 4 Project (P4P) is an authentic, research-based assessment that mirrors professional engineering practice. The P4P is aligned with the University of Auckland's Lane 2: Uncontrolled Assessment (Assessment FOR Learning) under the [Two Lane policy](#).

AI is therefore permitted, and may generally be used without restriction. However, students must follow University policies on [research conduct](#), including [data management](#). (Talk to your supervisor if you have questions around this.). Students may use generative AI tools to assist in creating assessment artefacts, and the submitted work is treated as the student's work.

AI tools have no agency in the assessment context. If you use an AI tool, you are treated as the author of any resulting text/code/analysis, and you are responsible for accuracy, limitations, referencing, and avoiding plagiarism. You must critically evaluate any AI output (including checking calculations, verifying claims against sources, and confirming technical correctness). Students may use AI tools for support activities such as:

- Planning and structure: outlining sections, improving report flow, producing checklists, drafting figure captions, or refining clarity and conciseness.
- Writing support: language editing, readability improvements, reducing repetition, and improving technical tone (without changing the underlying technical content you developed).
- Literature assistance: summarising papers you have read, generating keywords, helping compare/contrast methods, or turning notes into a structured narrative if you verify and cite primary sources.
- Technical support: creating and debugging code, explaining equations, assisting with data visualisation scripts.

Students may be required to discuss their intended use of AI tools with their supervisor and confirm that this discussion has taken place as part of their project plan submission in acknowledgment section.

9. Penalties and DNC Grades

Students are expected to follow professional standards and to submit all the compulsory project components on time and according to the provided instructions.

Final Report - Late Submission Penalty

All project components will be submitted through Canvas. Canvas automatically flags late submissions if they are submitted after the deadline even if your submission was uploaded a few seconds after the due date and time. Please make sure to submit at least 5 minutes before the deadline to avoid any late submission penalties. You can resubmit your files if you find a mistake has been made but note that only your latest submission will be considered. When resubmitting, you need to upload all the necessary files for that submission again. The following penalty will be applied for a late submission of the final report:

- **4% penalty per hour (or part hour) of lateness**

P4P Conference “No-Show” Penalties

Students are expected to participate in the P4P conference and give a seminar talk on their research. Students are also expected to attend **all sessions of the day** and attendance will be recorded at the beginning and the end of each session. If students fail to attend 2 or more sessions or present their work, this will incur a:

- **DNC grade** (Did Not Complete grade)

P4P Display Day “No-Show” Penalties

Students are expected to participate in the P4P display day and give a poster presentation of their research. Failure to participate and present the work will incur a:

- **DNC grade** (Did Not Complete grade)

10. Ethics Approval

Students conducting research involving human participants (such as interviews or surveys) are required to consult the guiding principles of the University of Auckland for conducting research with human participants. The Guidelines allow the University of Auckland to meet its obligation to ensure that all research by staff or students conforms to the highest appropriate ethical standards. In this way, research participants are treated with respect and dignity. Their privacy, safety, health, personal, social, and cultural sensitivities are protected. Those requiring University Ethics Approval should follow the guidelines and complete the research project application form that can be found at the following URL:

<http://www.auckland.ac.nz/uoa/re-uahpec>

11. Departmental P4P Team

Below you can find the names and roles of all the people involved in the departmental Final Year Projects team:

Dr. Maedeh Amirpour (Course Director)

Prof. Piaras Kelly (HOD)

Mr. Kevin Jia (Project Documentation)